

Case Study Review

I. Introduction

In *Ontology development for agriculture domain*, the authors propose the development of a generic ontology for categorizing relationships between agricultural data and knowledge reasoning rules, as no generic ontology applicable to the agricultural domain had previously existed (Malik, Sharan and Hijam, 2015). The authors assert that such an ontology would provide significant value to researchers as a means for more efficient text mining, and for increasing communicability of information between data-driven software systems, as well as between the human actors who work with them (Malik, Sharan and Hijam, 2015).

Building on the authors' conclusions, I further assert that such an ontology, if sufficiently developed, would also provide opportunities to detect and define both "known unknown" and "unknown unknown" categories of knowledge in a way that enhances its modularity, self-corrective ability, and practical usability.

II. Ontology Development Approaches

A. Types of Ontologies

Throughout their study, the authors rely on Gruber's definition of an ontology as a baseline conceptual guideline: "[a]n ontology is a formal, explicit specification of a shared conceptualization," or an abstract model for representing interrelated categories of information (Gruber, 2009; Malik, Sharan and Hijam, 2015). The authors then describe four broad types of ontologies:

1. Domain ontologies, used to define information about a particular subject matter domain;
2. Generic ontologies, designed to define information that applies to several subject matter domains;

3. Application ontologies, used to model all of the information and knowledge reasoning rules needed for a specific subject matter domain; and
4. Representational ontologies, designed to provide abstract structures for representing subject matter domains without stating or knowing what type of information is to be modeled (Malik, Sharan and Hijam, 2015).

The authors reference AGROVOC, a multilingual thesaurus of the Food and Agriculture Organization of the United Nations (AGROVOC, 2025; Malik, Sharan and Hijam, 2015). They observe that while AGROVOC contains a great deal of information, that information is ontologically unstructured and lacks a consistent means for accurately identifying relationships between interrelated sets of information; in that sense, while AGROVOC may be a useful dictionary of terms, it is not a fully-formed ontology (Malik, Sharan and Hijam, 2015).

B. Steps to Create an Ontology

The authors then list seven steps for creating a viable ontology for both the agricultural subject matter domain at hand, and in general:

1. Determine the ontology's scope within the chosen subject matter domain;
2. Evaluate any pre-existing ontologies for their applicability to the subject matter domain;
3. Identify and define the primary conceptual terms to be used in the new ontology;
4. Define classes of information within the ontology, and the relationships between classes;
5. Define the "slots," or properties of those same classes;
6. Define the "facets" of those same slots, or their details; and
7. Apply the new ontology in practice (Malik, Sharan and Hijam, 2015).

In terms of the agricultural subject matter domain, the authors landed on five main classes for their new ontology: Plant, Disease, Pest, Pesticide and Fertilizer (Malik, Sharan and Hijam, 2015), along with related subsets of subclasses. They emphasize that their approach is intended create a generic ontology that is amenable to refinement and to being expanded as new categories of information and

relationships between informational categories are discovered (Malik, Sharan and Hijam, 2015).

C. Use Cases for a Generic Ontology

The authors additionally assert several practical uses cases for generic ontologies such as theirs:

1. Automatic extraction of actionable knowledge (whether observationally-derived facts or reasoning rules) from texts;
2. Enhanced interoperability of database-driven systems, i.e., the standardization of knowledge representation within an ontology makes it more efficient for different software systems to intercommunicate;
3. Enhanced facilitation of communication between human actors in organizations working within the ontology's subject matter domain; and
4. Creation of a common knowledge base drawn from disparate sources that otherwise would be far more difficult or impossible to correlate in a usable manner (Malik, Sharan and Hijam, 2015).

III. Two Additional Practical Applications

I expand on the authors' proposal by asserting that two additional, and highly valuable, real world applications for such a generic ontology may be found by focusing on steps four through six for ontology creation, described above:

A. Identifying and Scoping Known Unknowns

A "known unknown" refers to that portion of a subject matter domain where researchers are aware in advance that gaps exist in the informational coverage available to them (Bi and Su, 2025; Grati, Fattouch and Boukadi, 2025). For example, researchers might be aware that a destructive fungus afflicts certain crops at specific stages of the crop's growth cycle, but they might not know its method of transmission between plants. For an ontology to be practically useful, that knowledge gap needs to be accounted for in relation to the well-known sections of the ontology (Grati, Fattouch and Boukadi, 2025).

B. Detecting Unknown Unknowns for Additional Research

An “unknown unknown” refers to that portion of a subject matter domain, or entire subject matter domains in and of themselves, where researchers don’t know what they don’t know, or don’t know what critical knowledge they might be lacking. (Bi and Su, 2025; Grati, Fattouch and Boukadi, 2025). For example, in the agricultural domain, researchers might have no awareness at all that a deleterious weed has the ability to invade tracts of cultivated crops, yet knowledge of the species would be critical for an ontology of those crops to be practically useful.

C. Developmental Approach

I further assert that one effective way for creating an ontology that is capable of detecting and defining both “known unknowns” and “unknown unknowns” would be to focus on categorical definitions that not only describe a concept or a fact representationally, but also include the expected, or even merely potential, interrelationships between those same categories as a baseline requirement. (Malik, Sharan and Hijam, 2015; Grati, Fattouch and Boukadi, 2025). Drawing from the authors’ criticism of AGROVAC, a set of ontological categories that defines types of weather (e.g., rain versus snow versus clear weather) would be purely representational on its face, in the sense that it simply asserts that such types of weather exist.

A more robust ontology would categorize types of weather based on their expected impact on a given type of crop, i.e., it would inherently contain the expected relationships with other categories of information in the ontological framework. For example, categories of rain might be established based on how likely they would be to help a crop’s growth versus how likely they would be to damage crops, such as rain that causes destructive flash flooding (Bi and Su, 2025).

IV. Business Context and Success Case Outcomes

As discussed above, the business context, or practical use cases, for investigating both “known unknowns” and “unknown unknowns” revolves around the goal of expanding the ontology’s utility in the real world.

Research of any kind can be time-consuming and resource-intensive; in the case of agricultural applications, there can be real world consequences for agricultural producers missing critical information about, for example, their expected crop yields in a given farming season (Grati, Fattouch and Boukadi, 2025; Lahoud et al, 2025). Such consequences have the potential to be quite severe, leading to outcomes such as food shortages or famines, loss of income or livelihoods to both the producers and the downstream supply chain of businesses that rely on predictable crop yields, and the social and political disruptions that may be unavoidable consequences of the same (Grati, Fattouch and Boukadi, 2025; Lahoud et al, 2025).

By developing self-correcting ontology with a deliberate focus on using it to determine what knowledge researchers are still missing, and where such knowledge might be found with further research, such catastrophic risks can be averted or ameliorated to the extent practical ahead of time (Lahoud et al, 2025).

In that sense, a successfully-implemented ontology needs to be both flexible and adaptive. It should not be viewed as a static system of purely representational definitions that never change once they have been created. Instead, it should not only be self-correcting, or capable of identifying the gaps in its informational coverage for its subject matter domain, but also be modular, or capable of incorporating new categories as new and relevant information is discovered (Grati, Fattouch and Boukadi, 2025; Lahoud et al, 2025).

V. Conclusion

Overall, I agree with the authors that generic ontologies provide multiple opportunities for real world use cases; I further assert that such ontologies will exhibit increased efficiency and higher value performance when they are designed from the start to be modular and self-correcting, meaning, amenable to detecting and remediating gaps in their informational coverage of their subject matter domains.

VI. References

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